

Christopher Alvarez Diaz

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PROFESSIONAL SUMMARY

Robotics Engineering student with a strong foundation in digital logic, RTL design, and systems integration. Hands-on experience developing and verifying a custom 8-bit CPU in Verilog using simulation tools, complemented by Python, Linux, and robotics development. Strong communicator with research presentation experience and an interest in hardware development and testing.

EDUCATION

University of California, Santa Cruz

Jun 2029

B.S., Robotics Engineering

TECHNICAL SKILLS

- **Hardware & Verification:** Verilog, RTL Simulation, AMD Vivado, Digital Logic, Computer Architecture
- **Software & Systems:** Python, Bash/Linux CLI, Git/GitHub, ROS 2, Nav2, AI, VSCode
- **Engineering Tools:** SolidWorks, OnShape, Gazebo Simulation

PROJECTS

Custom 8-Bit Single-Cycle CPU

Aug 2026

- Architected and implemented a complete 8-bit single-cycle processor from the gate level up, featuring a custom 12-instruction ISA with R/I/J-type encodings.
- Engineered the processor datapath, including the ALU, register file, control-unit decoder, and memory modules, with attention to critical-path efficiency.
- Tested the CPU with a comprehensive testbench suite, achieving a 100% pass rate in AMD Vivado behavioral simulations to verify logic operations and branch execution.

Autonomous ROS 2 Payload Drone

Mar 2026 - Present

- Developed an autonomous drone navigation software stack in Python, leveraging Nav2 and Vision AI for obstacle avoidance.
- Validated flight control algorithms through extensive simulated test flights across a virtual city environment in Gazebo prior to hardware deployment.
- Designed the physical drone platform using CAD tools, bridging the gap between software simulation and hardware execution.

Battery Sub-Team

Sep 2025 - Present

Formula Slug (EV Racing Team)

- Researched and designed bus bar configurations and actuator systems for a high-performance 120V EV battery pack using SolidWorks and OnShape & McMaster Carr. Tested with oscilloscopes and multi meters.
- Optimized the battery layout architecture, successfully reducing the total size of the battery tray by 60%.
- Assembled and integrated low/high-voltage components, achieving a 100% pass rate during vehicle electrical testing.

PROFESSIONAL EXPERIENCE

Antelope Valley College

Oct 2024 - Jun 2025

Undergraduate Biology Researcher

- Conducted original research on micro-arthropod behaviors in controlled environments, becoming the first high school student in the region to complete independent research in collaboration with the college.
- Analyzed and presented research findings at the Spring Research Symposium and to a class of 32 students; the research is currently in development for academic publication.

UC Santa Cruz

Jan 2026 - Present

Administrative and Financial Intern

- Process and track approximately 150 campus business forms, including requisitions and vendor registrations, each month while maintaining 100% compliance with university financial policies.
- Execute detailed budget projections for multiple departments to analyze resource allocation and improve financial efficiency.